

FINAL TEST SERIES for NEET-2022

MM : 720

Test - I2

Time : 3 Hrs. 20 Mins.

Answers

1. (1)	41. (3)	81. (4)	121. (1)	161. (3)
2. (1)	42. (1)	82. (3)	122. (2)	162. (3)
3. (2)	43. (2)	83. (1)	123. (4)	163. (3)
4. (4)	44. (4)	84. (4)	124. (3)	164. (2)
5. (2)	45. (4)	85. (3)	125. (1)	165. (2)
6. (1)	46. (2)	86. (3)	126. (4)	166. (3)
7. (4)	47. (1)	87. (1)	127. (4)	167. (4)
8. (1)	48. (4)	88. (4)	128. (3)	168. (2)
9. (1)	49. (2)	89. (3)	129. (1)	169. (3)
10. (4)	50. (1)	90. (1)	130. (1)	170. (2)
11. (3)	51. (2)	91. (2)	131. (2)	171. (3)
12. (2)	52. (4)	92. (3)	132. (2)	172. (3)
13. (2)	53. (4)	93. (4)	133. (4)	173. (3)
14. (4)	54. (3)	94. (4)	134. (1)	174. (1)
15. (4)	55. (2)	95. (3)	135. (2)	175. (2)
16. (2)	56. (4)	96. (2)	136. (2)	176. (1)
17. (2)	57. (1)	97. (4)	137. (3)	177. (4)
18. (2)	58. (3)	98. (2)	138. (2)	178. (2)
19. (2)	59. (4)	99. (3)	139. (3)	179. (4)
20. (3)	60. (3)	100. (3)	140. (1)	180. (4)
21. (2)	61. (1)	101. (3)	141. (3)	181. (4)
22. (4)	62. (4)	102. (2)	142. (3)	182. (2)
23. (2)	63. (3)	103. (4)	143. (4)	183. (1)
24. (2)	64. (4)	104. (2)	144. (2)	184. (4)
25. (2)	65. (3)	105. (4)	145. (1)	185. (3)
26. (1)	66. (3)	106. (4)	146. (1)	186. (2)
27. (1)	67. (1)	107. (3)	147. (2)	187. (2)
28. (1)	68. (4)	108. (2)	148. (3)	188. (2)
29. (4)	69. (3)	109. (3)	149. (4)	189. (3)
30. (3)	70. (2)	110. (3)	150. (3)	190. (1)
31. (4)	71. (1)	111. (3)	151. (3)	191. (2)
32. (1)	72. (3)	112. (3)	152. (2)	192. (3)
33. (3)	73. (1)	113. (3)	153. (4)	193. (2)
34. (4)	74. (1)	114. (4)	154. (3)	194. (4)
35. (2)	75. (3)	115. (1)	155. (2)	195. (1)
36. (1)	76. (2)	116. (3)	156. (3)	196. (3)
37. (2)	77. (2)	117. (4)	157. (3)	197. (2)
38. (2)	78. (2)	118. (2)	158. (3)	198. (4)
39. (4)	79. (3)	119. (2)	159. (2)	199. (1)
40. (4)	80. (3)	120. (1)	160. (2)	200. (4)

FINAL TEST SERIES for NEET-2022

MM : 720

Test - 12

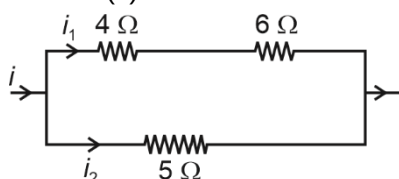
Time : 3 Hrs. 20 Mins.

Hints and Solutions

PHYSICS

SECTION - A

1. Answer (1)



$$\text{Given } i_2^2 \times 5 = 20$$

$$i_2 = 2 \text{ A}$$

$$\frac{i_1}{i_2} = \frac{5}{10} = \frac{1}{2}$$

$$i_1 = 1 \text{ A}$$

$$P_{4\Omega} = i_1^2 \times 4 = 1^2 \times 4 = 4 \text{ J/s}$$

Heat generated through 4Ω resistor is equal to
 $H = 4 \times 1 = 4 \text{ J}$

2. Answer (1)

In series resonance circuit, both L and C are present in the circuit.

At resonance V_L and V_C have 180° phase difference and total source voltage appears across R .

3. Answer (2)

Intensity of EM wave is given by $I = \epsilon_0 E_{rms}^2 c$

$$I = 8.85 \times 10^{-12} \times 4 \times 10^4 \times 3 \times 10^8 \\ = 106.2 \text{ W/m}^2$$

4. Answer (4)

For image to be coincide with object, it should be placed on centre of curvature of the effective mirror.

$$\text{i.e. } d = 2 |f_{\text{eff}}|$$

Now,

$$\frac{1}{f_{\text{eff}}} = \frac{-2}{f_l} + \frac{1}{f_m} = \frac{-2}{10} + \frac{1}{\infty}$$

$$f_{\text{eff}} = -5 \text{ cm}$$

$$d = 2 \times 5 = 10 \text{ cm}$$

5. Answer (2)

$$m = \frac{D}{f_e} = \frac{25}{5} = 5$$

6. Answer (1)

From doppler's shift

$$\frac{\Delta\lambda}{\lambda} = \frac{v_{\text{rad}}}{c}$$

When source moves away from observer, the frequency measured will be smaller and wavelength will be longer. This is called red shift.

7. Answer (4)

From voltage divider rule

$$V_{R_2} = \frac{4}{4+5} \times 12 = 5.33 \text{ V}$$

Since $V_z > V_{R_2} \Rightarrow$ zener diode will remain off

$$\text{And } V_0 = V_{R_2} = 5.33 \text{ V}$$

8. Answer (1)

$$E_p = \frac{hc}{\lambda}$$

$$E_p = \frac{-13.6}{16} - \left(-\frac{13.6}{1} \right) = 12.75 \text{ eV}$$

$$\lambda = \frac{12400}{12.75} \text{ \AA}$$

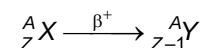
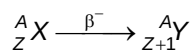
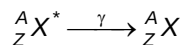
$$\lambda = 972.54 \text{ \AA}$$

9. Answer (1)

$$1 \text{ curie} = 3.7 \times 10^{10} \text{ decay/s}$$

$$\text{or } 1 \text{ curie} = 3.7 \times 10^{10} \text{ Bq}$$

10. Answer (4)



11. Answer (3)

$$E_{\text{centre}} = \frac{2K\lambda}{R} (-\hat{j})$$

12. Answer (2)

$$I = \frac{15 - 0.3}{(10 + 20)} = \frac{14.7}{30}$$

$$= 0.49 \text{ A}$$

13. Answer (2)

$$y = 10 \sin\left(\frac{2\pi}{3}x\right) \cos(100\pi t)$$

$$\Rightarrow \lambda = 3 \text{ cm and } f = 50 \text{ Hz}$$

Now separation between two adjacent antinodes is $\frac{\lambda}{2}$

$$\Rightarrow d = \frac{\lambda}{2} = \frac{3}{2} \text{ cm}$$

14. Answer (4)

$$\frac{1}{\epsilon_0} \frac{e^2 r}{hc} = \frac{Fr^3}{\lambda E} = \frac{[M L T^{-2}][L^3]}{[L][M L^2 T^{-2}]} = [M^0 L T^0 A^0]$$

15. Answer (4)

$$\% \text{ loss of energy} = \frac{\Delta U}{U} \times 100\%$$

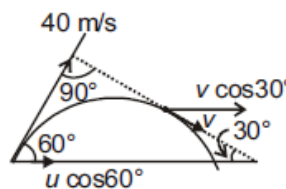
$$\frac{\Delta U}{U} \times 100\% = \frac{\frac{C_1 C_2}{2(C_1 + C_2)} (V_1 - V_2)^2}{\frac{1}{2} C_1 V^2} \times 100\%$$

$$= \frac{2 \times 8}{2 \times 10} V^2 \times 100\%$$

$$= \frac{1}{2} \times 2 \times V^2$$

$$= 80\%$$

16. Answer (2)



When velocity is perpendicular to initial velocity

$$v \cos 30^\circ = u \cos 60^\circ$$

$$v \times \frac{\sqrt{3}}{2} = 40 \times \frac{1}{2}$$

$$v = \frac{40}{\sqrt{3}} \text{ m/s}$$

17. Answer (2)

$$\text{Resultant force } F = Kt^2 - \mu mg$$

$$a = \frac{k}{m} t^2 - \mu g$$

It is equation of parabola

18. Answer (2)

Work done by resultant force = Change in kinetic energy

$$v = a\sqrt{x}$$

$$v^2 = a^2 x$$

$$2v \frac{dv}{dt} = \frac{a^2}{2} v$$

$$\int_0^v dv = \frac{a^2}{2} \int_0^t dt$$

$$v = \frac{a^2}{2} t$$

$$\Delta K = \frac{1}{8} m a^4 t^2$$

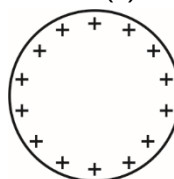
19. Answer (2)

$$K.E = \frac{L^2}{2I}$$

20. Answer (3)

$$V = f\lambda$$

21. Answer (2)



Due to repulsion, radius increases

22. Answer (4)

$$\text{Force on side } PR = iBL \cos 30^\circ$$

$$= iBL \frac{\sqrt{3}}{2}$$

23. Answer (2)

$$\frac{1}{\lambda} = RZ^2 \left[\frac{1}{2^2} - \frac{1}{3^2} \right]$$

$$\frac{1}{x} = RZ^2 \left[\frac{1}{1^2} - \frac{1}{2^2} \right]$$

$$\therefore \frac{x}{\lambda} = \frac{5}{\frac{36}{3}} = \frac{5}{27}$$

$$\therefore x = \frac{5\lambda}{27}$$

24. Answer (2)

$$t = \frac{\sqrt{v_1 v_2}}{g} = \frac{\sqrt{20 \times 20}}{10} = 2 \text{ s}$$

$$x = 40 \times 2 = 80 \text{ m.}$$

25. Answer (2)

$$a = \frac{(60 - 15 - 10 - 15)}{2 + 3} = 4 \text{ m/s}^2$$

$$60 - T - 15 = 3 \times 4$$

$$\therefore T = 33 \text{ N}$$

26. Answer (1)

$$a = \frac{2g}{3} = \frac{2 \times 10}{3} = 6.6 \text{ m/s}^2$$

$$\alpha = \frac{a}{R} = \frac{100}{3} \text{ rad/s}^2$$

27. Answer (1)

$$V_x = 40 \text{ volt}$$

$$V_y = 32 \text{ volt}$$

$$\therefore V_x - V_y = 8 \text{ volt}$$

28. Answer (1)

$$M = \sqrt{L_1 L_2}$$

$$\text{EMF} = M \frac{di}{dt} = \frac{(6 \times 10^{-3})(4)}{2 \times 10^{-3}} = 12 \text{ volt}$$

29. Answer (4)

$$\text{Maximum tension in string } T = m(g + a)$$

$$= 1000 (9.8 + 2.76)$$

$$= 12.56 \times 10^3 \text{ N}$$

$$\text{B.S} = \frac{T}{A} = \frac{12.56 \times 10^3}{\pi r^2} = 1.44 \times 10^9$$

$$D = 2r = 3.33 \text{ mm}$$

30. Answer (3)

Cylinder is in unstable equilibrium when vertical

31. Answer (4)

$$v \propto r^2$$

$$\left(\frac{1}{9} \right) = \left(\frac{r}{R} \right)^2$$

$$R = 3r$$

$$\frac{M}{m} = \left(\frac{R}{r} \right)^3$$

32. Answer (1)

$$\text{Actual efficiency } \eta_1 = \frac{1.2}{4.2} = \frac{2}{7}$$

$$\text{Theoretical efficiency } \eta_2 = 1 - \frac{T_2}{T_1} = \frac{3}{8}$$

$$\text{Required ratio} = \frac{16}{21}$$

33. Answer (3)

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

34. Answer (4)

$$\chi_m = \frac{I}{H} = 5000$$

$$\mu_r = 1 + \chi_m$$

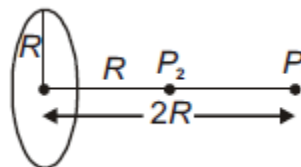
$$= 5001$$

35. Answer (2)

$$230 - (112 + 106) = 12 \text{ MeV}$$

SECTION - B

36. Answer (1)



$$V_{P_1} = -\frac{GM}{R\sqrt{5}}$$

$$V_{P_2} = -\frac{GM}{R\sqrt{2}}$$

$$-m[V_{P_2} - V_{P_1}] = \frac{1}{2}mv^2$$

$$= (V_{P_1} - V_{P_2})m = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2GM}{R\sqrt{2}} \left(1 - \sqrt{\frac{2}{5}} \right)}$$

37. Answer (2)

$$\vec{E}_x = -\frac{dV}{dx} = -[6 - 8y^2] \hat{i}$$

$$\vec{E}_y = -\frac{dV}{dy} = -[-16xy - 8 + 6z] \hat{j}$$

$$\vec{E}_z = -\frac{dV}{dz} = -[6y - 8z] \hat{k}$$

at origin $\vec{E} = -6\hat{i} + 8\hat{j} + 0\hat{k}$

$$|\vec{E}| = 10 \frac{\text{N}}{\text{C}}$$

$$F = qE = 20 \text{ N}$$

38. Answer (2)

$$i = \frac{10}{10} = 1 \text{ A}$$

$$\therefore P = i^2 R = 1^2 \times 10 = 10 \text{ W}$$

39. Answer (4)

$$(x+1)520 = (x)780$$

$$\Rightarrow x = \frac{520}{260} = 2$$

40. Answer (4)

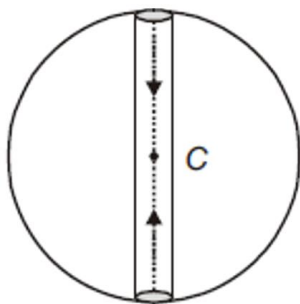
$$dQ = mc dt$$

$$dQ = 10 \times 0.3 t^2 dt$$

$$Q = 3 \int_0^{10} t^2 dt$$

$$= 1000 \text{ cal}$$

41. Answer (3)



$$T = 2\pi \sqrt{\frac{R}{g}}$$

42. Answer (1)

$$v_f = 5 + (\text{area under } a - t) = -2.5 \text{ m/s}$$

43. Answer (2)

5 kg block is slipping

$$a = \frac{(F_1 - F_2) - 5g\mu}{5} = 3 \text{ m/s}^2$$

$$\therefore \text{contact force} = 10 + 3 \times 2 = 16 \text{ N}$$

44. Answer (4)

$$v = gt$$

$$\frac{\Delta g}{g} \times 100 = \frac{\Delta v}{v} \times 100 + \frac{\Delta t}{t} \times 100$$

$$\frac{\Delta g}{g} \times 100 = 2 + 2 = 4\%$$

45. Answer (4)

$$v_A = \frac{dx_A}{dt} = 3 + 2t$$

$$v_B = \frac{dx_B}{dt} = 4t - 1$$

$$\text{then } v_A = v_B$$

$$3 + 2t = 4t - 1$$

$$4 = 2t$$

$$t = 2 \text{ s}$$

46. Answer (2)

$$T = \frac{2u_y}{g}$$

$$10 = \frac{2u_y}{g}$$

$$H = \frac{u_y^2}{2g} = \frac{1}{2} \left(\frac{u_y}{g} \right)^2 g = \frac{1}{2} (5)^2 \times 10$$

$$= 125 \text{ m}$$

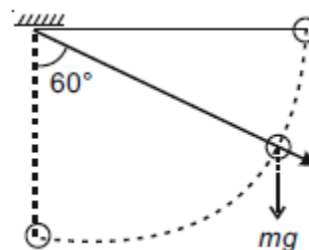
47. Answer (1)

$$a = \frac{(5 - 3 - 2\mu_k)g}{(5 + 3 + 2)}$$

$$1 \times 10 = (2 - 2\mu_k)10$$

$$\text{then } \mu_k = 0.5$$

48. Answer (4)



$$T - mg \cos 60^\circ = \frac{mv^2}{l}$$

$$v^2 = 2gl(1 - \cos 60^\circ)$$

$$v^2 = gl$$

$$T = \frac{mg}{2} + mg$$

$$T = \frac{3}{2}mg$$

49. Answer (2)

$$y = A + \overline{A \cdot B} = A + (\overline{A} + \overline{B})$$

$$= (A + \overline{A}) + \overline{B}$$

$$= 1 + \overline{B}$$

$$= 1$$

50. Answer (1)

Total energy of SHM is constant. At any instant

$$E = \frac{1}{2}kx^2 + \frac{1}{2}I\omega^2 + \frac{1}{2}mv^2$$

$$E = \frac{1}{2}kx^2 + \frac{3}{4}mv^2$$

$$\frac{dE}{dt} = \frac{1}{2}k \cdot 2x \frac{dx}{dt} + \frac{3}{4}m \cdot 2v \frac{dv}{dt}$$

$$0 = kxv + \frac{3}{2}mav$$

$$a = -\left(\frac{2k}{3m}\right)x$$

$$\Rightarrow \omega^2 = \frac{2k}{3m}$$

$$\omega = \sqrt{\frac{2k}{3m}}$$

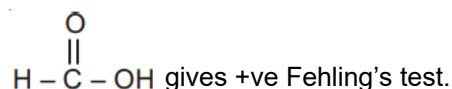
$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{2 \times 300}{3 \times 2}} = \frac{10}{2\pi}$$

$$f = \frac{5}{\pi} \text{ Hz}$$

CHEMISTRY

SECTION - A

51. Answer (2)



52. Answer (4)

In gaseous state, 3° amine is strongly basic.

53. Answer (4)

Gold is a negatively charged sol. So AlCl_3 is most effective.

54. Answer (3)

Br attach to benzene ring will not participate in S_N reaction due to double bond character between C-Br bond.

55. Answer (2)

Bakelite is a phenol formaldehyde resin.

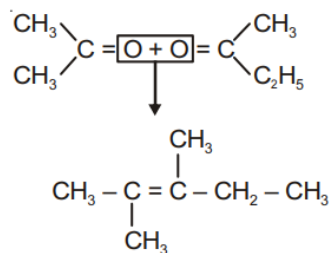
56. Answer (4)

 $\text{CO} + \text{N}_2$ is producer gas.

57. Answer (1)

Calgon is chemically sodium hexametaphosphate.

58. Answer (3)



2, 3-Dimethyl pent-2-ene.

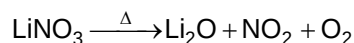
59. Answer (4)

Fructose contains ketonic group while glucose contains aldehydic group.

60. Answer (3)

For isoelectronic species, size increases with decrease in atomic numbers.

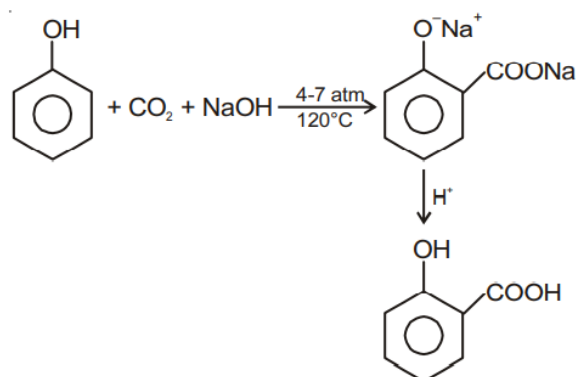
61. Answer (1)



62. Answer (4)

Kolbe electrolysis used to prepare hydrocarbons only.

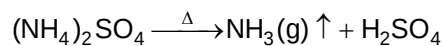
63. Answer (3)



64. Answer (4)

Aldehydes without α -hydrogens undergo cannizzaro reaction.

65. Answer (3)



66. Answer (3)



67. Answer (1)

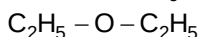
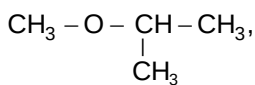
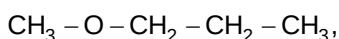
In pyrosilicate two units share one oxygen atom.

68. Answer (4)

Contact process is used to manufacture H_2SO_4 .

69. Answer (3)

Three metamers are possible



70. Answer (2)

At high pressure, $PV = RT + Pb$

$$\therefore \frac{PV}{RT} = 1 + \frac{Pb}{RT}$$

$$\therefore Z = 1 + \frac{Pb}{RT}$$

71. Answer (1)

$$E = \frac{3}{2}RT = \frac{3}{2}PV$$

$$P = \frac{2E}{3V} = \frac{2E}{3} \quad [\because V = 1\text{L}]$$

72. Answer (3)

Osmosis takes place from B to A as the osmotic pressure of A is higher than B.

73. Answer (1)

n-factor = 1

74. Answer (1)

Ratio of closed packed atoms to octahedral voids is 1 : 1.

75. Answer (3)

Since reference state of carbon is graphite therefore ΔH_f° of graphite is zero.

76. Answer (2)

Heat liberated by the formation of 18 g $\text{H}_2\text{O} = 36 \text{ kJ}$

$$\text{Heat liberated by the formation of 1 g } \text{H}_2\text{O} = \frac{36}{18}$$

$$= 2 \text{ kJ.}$$

77. Answer (2)

$$\text{As } Q_c = \frac{(2)^2}{2 \times 2} = 1 \text{ as } Q_c > K_c$$

So reactions will shift in the backward direction

78. Answer (2)

More negative E_{SRP}° , more stronger is the reducing agent.

79. Answer (3)

Due to very less probability of simultaneous collision of more than three reacting species at a time, reaction with order greater than three are rare.

80. Answer (3)

$$\therefore \text{No. of moles} = \frac{10}{20} = 1/2$$

$$\therefore \text{No. of molecules} = \frac{N_A}{2}$$

$$\begin{aligned} \therefore \text{No. of electrons} &= 10 \times \frac{N_A}{2} \\ &= 5 N_A \end{aligned}$$

81. Answer (4)

For $l = 2$,

$$\begin{aligned} \text{Orbital angular momentum} &= \hbar \sqrt{l(l+1)} \\ &= \hbar \sqrt{2(2+1)} \\ &= \sqrt{6} \hbar \end{aligned}$$

For $l = 3$,

$$\begin{aligned} \text{Orbital angular momentum} &= \hbar \sqrt{3(3+1)} \\ &= \hbar \sqrt{12} \end{aligned}$$

$$\therefore \text{Ratio} = \sqrt{6} : \sqrt{12} = 1 : \sqrt{2}$$

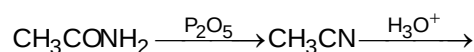
82. Answer (3)

$$\text{Amount converted to product} = x - \frac{x}{8} = \frac{7x}{8}$$

83. Answer (1)

Value of $m = -l$ to $+l$

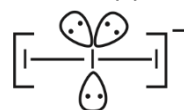
84. Answer (4)



85. Answer (3)

Intermediate is CCl_2 in both the reactions.**SECTION - B**

86. Answer (3)



87. Answer (1)

Pine oils, fatty acids and xanthates are used as collectors in froth floatation.

88. Answer (4)

The order of oxidising power is based on the increasing stability of the lower species to which they are reduced.

89. Answer (3)

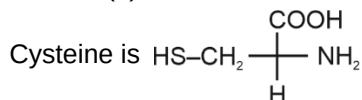
$$t_{99.9} = \frac{2.303}{k} \log \frac{100}{100 - 99.9} \quad \dots (1)$$

$$t_{99} = \frac{2.303}{k} \log \frac{100}{100 - 99} \quad \dots (2)$$

Using equations (1) & (2)

$$\frac{t_{99.9}}{t_{99}} = \frac{3}{2}$$

90. Answer (1)



91. Answer (2)

If $\Delta S < 0$ and $\Delta H > 0$ then ΔG of the process will be always positive at all temperature therefore always be non spontaneous.

92. Answer (3)

Species Bond order Magnetic nature

O_2 2 Paramagnetic

O_2^+ 2.5 Paramagnetic

O_2^- 1.5 Paramagnetic

O_2^{2-} 1 Diamagnetic

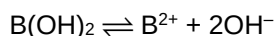
N_2 3 Diamagnetic

N_2^- 2.5 Paramagnetic

C_2 2 Diamagnetic

C_2^- 2.5 Paramagnetic

93. Answer (4)



(i) — — 10^{-2}

(eq) — S $2S + 10^{-2}$

$$K_{sp} = [\text{B}^{2+}] [\text{OH}^-]^2$$

$$= (S) (2S + 10^{-2})^2 \quad [\because 2S \ll 10^{-2} \therefore 2S + 10^{-2} \approx 10^{-2}]$$

$$10^{-8} = (S) (10^{-2})^2$$

$$S = 10^{-4} \text{ M}$$

94. Answer (4)

$$\therefore \text{Un} = 1, \quad \text{nil} = 0 \quad \text{pent} = 5.$$

$\therefore$ IUPAC name of element having $Z = 105$ is Unnilpentium.

95. Answer (3)

Using Kohlrausch's law

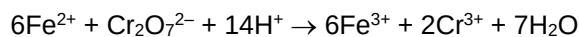
$$\Lambda_E^\circ (\text{CH}_3\text{COOH}) = \Lambda_E^\circ (\text{CH}_3\text{COO})_2\text{Ca} + \Lambda_E^\circ \text{HCl} - \Lambda_E^\circ \text{CaCl}_2$$

$$= z + y - x$$

96. Answer (2)

Maximum prescribed concentration of Fe in drinking water is 0.2 ppm

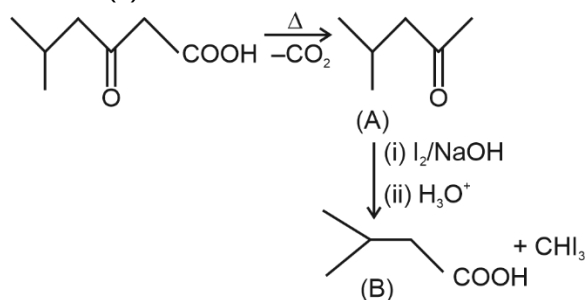
97. Answer (4)



$\therefore$ 1 mol $\text{Cr}_2\text{O}_7^{2-}$ can oxidise 6 mol Fe^{2+} ions

$\therefore$ 2 mol $\text{Cr}_2\text{O}_7^{2-}$ can oxidise 12 mol Fe^{2+} ions

98. Answer (2)



99. Answer (3)

H_2O has lone pairs therefore it is an electron rich hydride.

100. Answer (3)

$\text{B}_3\text{N}_3\text{H}_6$ has structure similar of benzene therefore known as inorganic benzene.

BOTANY

SECTION - A

101. Answer (3)

Pili in bacteria is usually involved in partial transfer of DNA from one cell to another cell.

102. Answer (2)

- Prokaryotic cells lack membrane-bound organelles.
- Ribosomes are non-membrane bound organelles found in both eukaryotic as well as prokaryotic cells

103. Answer (4)

- In telophase, nuclear envelope develops around the chromatin clusters at each pole forming two daughter nuclei.
- Chromosomes appear like a ball of wool in prophase.

104. Answer (2)

Museum is a place used for storing, preservation and exhibition of both plants and animals.

105. Answer (4)

Consciousness is a defining property of all living organisms.

106. Answer (4)

Glycolysis is also called EMP pathway.

The pacemaker enzyme (PFK) in glycolysis catalyses the conversion of Fructose-6-phosphate to Fructose-1, 6-bisphosphate, that is a rate limiting step of glycolysis.

107. Answer (3)

One molecule of GTP is formed during conversion of succinyl CoA to succinic acid in Krebs' cycle.

108. Answer (2)

Strobilanthes kunthiana flowers once in 12 years.

109. Answer (3)

Isogametes are produced in some algae and fungi.

110. Answer (3)

Pollen grains are surrounded by mucilaginous covering in flowers that are pollinated by water.

111. Answer (3)

Genotype of plants which are homozygous for both the traits in F_2 generation w.r.t Mendelian dihybrid cross:

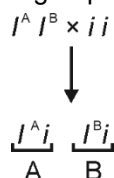
1RRYY, 1rryy, 1RRyy, 1rrYY

Therefore, among 16 individuals: 4 are homozygous for both the traits in F_2 generation.

$$= \frac{4}{16} \times 100 \Rightarrow 25\%$$

112. Answer (3)

Parents with genotype $I^A I^B$ and $i i$ cannot produce progeny with blood group 'O'.



113. Answer (3)

Round seed and inflated pod are dominant traits. Therefore, they express in both homozygous as well as heterozygous condition.

114. Answer (4)

DNA ligase is employed during replication on template with polarity $5' \rightarrow 3'$ where discontinuous replication occurs.

115. Answer (1)

During Griffith's experiment:

- When S-strain injected into mice, mice died
- When S-strain (heat-killed) + R strain (live) injected into mice, mice died.

116. Answer (3)

Lac y gene in Lac operon codes for permease that increases permeability of cell to β -galactosides.

117. Answer (4)

Saccharum barberi had low sugar content. *Saccharum officinarum* was originally grown in south India.

118. Answer (2)

Pusa Sawani is resistant to shoot and fruit borer.

119. Answer (2)

Mortality is the number of deaths in the population during a given period.

120. Answer (1)

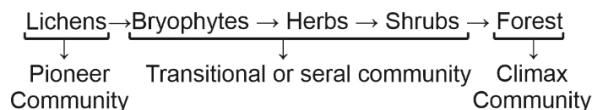
Logistic growth describes a situation when resources present in the environment are limiting.

121. Answer (1)

In competition and amensalism, none of the interacting species are benefitted.

122. Answer (2)

Sequence of various stages in xerarch succession is as:



123. Answer (4)

Reservoir pool is earth's crust or lithosphere for phosphorous.

124. Answer (3)

Pyramid of energy is always upright.

125. Answer (1)

Dobson unit (DU) measures thickness of ozone in a column from ground to top of atmosphere.

126. Answer (4)

DDT is found to be in highest concentration at top consumer level or highest trophic level.

127. Answer (4)

If the pressure greater than atmospheric pressure is applied to pure water or a solution, its water potential increases.

128. Answer (3)

Facilitated transport requires special membrane proteins. It is ATP independent.

129. Answer (1)

Mg^{2+} is an activator for both RuBisCO and PEPCase, both of which are critical enzymes involved in carbon fixation during photosynthesis.

130. Answer (1)



131. Answer (2)
Dikaryophase is found in basidiomycetes and ascomycetes.
132. Answer (2)
Keel is characteristic feature of Fabaceae family.
133. Answer (4)
Thermoacidophiles are chemoautotrophic and facultative anaerobes.
134. Answer (1)
Deuteromycetes are imperfect fungi. Fruiting bodies is absent in their life cycle.
135. Answer (2)
Cells in elongation region are responsible for growth of root in length.

SECTION - B

136. Answer (2)
Teak shows inflorescence of Cymose type. It has flowers borne in basipetal succession.
137. Answer (3)
Pericycle is parenchymatous in dicot root.
138. Answer (2)
Strobilus is found in – *Equisetum*, *Selaginella* and Gymnosperms (*Pinus*).
139. Answer (3)
Rice is C_3 plant. It has RuBisCO as primary CO_2 acceptor
140. Answer (1)
Triple response is shown by Ethylene
Cell division is stimulated by plants cytokinin

- Bolting in rosette plants is stimulated by gibberellins
141. Answer (3)
In a typical angiospermic plant,
Integument – diploid (2n)
Antipodal cell – haploid (n)
142. Answer (3)
Pyrenoids store protein and starch.
143. Answer (4)
Equisetum is a pteridophyte. All of the given features are seen in *Equisetum*.
144. Answer (2)
Amides are formed by addition of another amino group to amino acids
145. Answer (1)
NAA is synthetic auxin, 2,4-D is a weedicide.
146. Answer (1)
Flocs are masses of aerobic bacteria associated with fungal filaments.
147. Answer (2)
Rice crop fields are man made ecosystem.
148. Answer (3)
Guanine is a purine
149. Answer (4)
Anabaena is used as biofertilizer
150. Answer (3)
Maximum biodiversity is contributed by angiosperms w.r.t plant global biodiversity.

ZOOLOGY**SECTION - A**

151. Answer (3)
Role of O_2 in the regulation of respiratory rhythm is quite insignificant.
152. Answer (2)
Earthworm is an annelid with closed circulatory system.
153. Answer (4)
All changes that occur are due to activation of fight and flight response due to sympathetic stimulation.
Micturition is inhibited in sympathetic stimulation.
154. Answer (3)
Patella is the largest sesamoid bone in the body formed by ossification within the quadriceps tendon.

155. Answer (2)
Myxine (Hag fish) is a hermaphrodite and exclusively marine.
Petromyzon is anadromous and development is indirect through ammocoete larva.
156. Answer (3)
Desired gene replicates alongwith cloning vector inside the host cells.
157. Answer (3)
The C peptide is not present in mature insulin and is removed during maturation into insulin.
158. Answer (3)
Brain stem = mid brain + pons + medulla
159. Answer (2)
Dorsal hollow nerve cord differentiates into brain and spinal cord. It is present above the notochord.

160. Answer (2)

Vinblastin is an alkaloid obtained from a plant called *Catharanthus roseus*. It is an important drug used as a chemotherapeutic agent for treating leukaemia.

161. Answer (3)

Fishes and amphibians are anamniotes. Reptiles, birds and mammals are amniotes.

162. Answer (3)

Dolly was the first mammal to be successfully cloned but there was no genetic modification carried out on adult donor nucleus. Therefore, it is not a transgenic animal. Polly and Molly were clones and transgenic animals.

163. Answer (3)

Parotid glands are absent in frogs (e.g. *Rana*), whereas present in toads (e.g., *Bufo*).

164. Answer (2)

Sucrose = Glucose + Fructose

It is a non-reducing sugar because both, aldehyde group of glucose and ketone group of fructose participate in the formation of glycosidic bond between the two.

165. Answer (2)

High level of progesterone helps to maintain endometrium and prevents menstrual bleeding.

Pitocin is synthetic oxytocin given to induce labor.

166. Answer (3)

Downstream processing is the name given to the stages after fermentation when the desired product is recovered and purified.

167. Answer (4)

Widal test – Typhoid

Schick test – Diphtheria

Gravidex test – Pregnancy

Mantoux test – Tuberculosis

168. Answer (2)

70% of CO₂ is carried as bicarbonate

7% of CO₂ is carried in dissolved state.

169. Answer (3)

The elevated calcium ions concentration causes exocytosis of synaptic vesicles.

170. Answer (2)

Plasmodium reproduces asexually in RBCs of man. Sexual cycle occurs in *Anopheles* mosquito but gametocytes are formed within RBCs.

171. Answer (3)

Africa is called as 'Cradle of Life' as the primitive bipedal ancestors of man arose in this continent.

172. Answer (3)

Protonephridia are primarily concerned with ionic and fluid volume regulation *i.e.* osmoregulation. It developed as a defence mechanism which copes with a hypotonic environment by removing excess water from an organism.

173. Answer (3)

The muscle bundles are bounded together by a common collagenous sheath of connective tissue called fascia.

174. Answer (1)

Platyhelminthes are triploblastic acoelomates whereas Aschelminthes are triploblastic pseudocoelomates.

175. Answer (2)

Hemozoin is formed within RBCs and is responsible for paroxysm of malaria.

176. Answer (1)

Nereis belongs to phylum Annelida. *Chaetopleura* and *Octopus* belong to Mollusca and *Limulus* is an arthropod.

177. Answer (4)

Pneumatic bones of birds provide buoyancy.

178. Answer (2)

Intervertebral disc is formed by fibrous cartilage.

179. Answer (4)

Thrombin is responsible for transformation of fibrinogen into fibrin clot.

180. Answer (4)

Skeletal muscle fibres show structural syncytium. Functional syncytium is shown by cardiac muscle fibres. In a meromyosin, head and short arm together constitutes cross-arm.

181. Answer (4)

The exaggerated response of the immune system to antigens is called allergy.

182. Answer (2)

Bilirubin is a bile pigment present in bile juice. Chylomicrons are absorbed and transported through lacteals. Salivary glands are situated outside the buccal cavity.

183. Answer (1)

Angina is acute chest pain due to reduced blood circulation to cardiac muscles. Persons with AB blood group are universal recipients.

184. Answer (4)

Both proximal convoluted tubule (PCT) and distal convoluted tubule (DCT) are present in renal cortex, outside the medullary pyramids.

185. Answer (3)

TV = 500 mL, residual volume = 1200 mL

Expiratory reserve volume = 1100 mL

Vital capacity = 4600 mL

So, $a < c < b < d$

SECTION - B

186. Answer (2)

Ball and socket joint – Between humerus and glenoid cavity of pectoral girdle

Hinge joint – Between humerus and radius, ulna

Saddle joint – Between carpal and metacarpal of thumb

187. Answer (2)

Depolarisation occurs due to opening of voltage gated Na^+ channels, whereas repolarisation occurs due to opening of voltage gated K^+ channels and closure of voltage gated Na^+ channels.

188. Answer (2)

Vestibular apparatus consists of semi-circular canals, utricle and saccule. Otolith organ consists of only utricle and saccule.

189. Answer (3)

Tetany is caused due to hyposecretion of parathormone; Diabetes mellitus is caused due to deficiency of insulin; Dwarfism is caused due to deficiency of somatotropin (GH); and cretinism is caused due to deficiency of thyroxine.

190. Answer (1)

Thyroxine increases BMR of the body and has calorogenic effect (It increases body temperature).

191. Answer (2)

Chromosome number in meiocytes of rat and cat are 42 and 38 respectively.

192. Answer (3)

The mitotic division starts as the zygote moves through the isthmus of oviduct towards uterus.

193. Answer (2)

Zona pellucida is synthesized by secondary oocyte. The cyclic events starting from one menstruation to the next is called menstrual cycle. Secretion of estrogen starts within growing follicle.

194. Answer (4)

GIFT involves *in vivo* fertilization.

195. Answer (1)

Dengue and chikungunya are viral diseases transmitted by female *Culex* mosquito. Filariasis is caused by an aschelminth and malaria is a disease caused by a protozoan. Small pox is a viral disease transmitted through droplets.

196. Answer (3)

Darwin's finches were small and black coloured birds and evolved from seed eating finches.

197. Answer (2)

Gene flow includes immigration and emigration which leads to change in allele frequencies. Mutation is source of new allele. Natural selection depends upon differential reproduction among various individuals.

198. Answer (4)

In MOET, 8-32 cells stages of embryo are recovered non-surgically and transferred to surrogate mothers.

199. Answer (1)

Bioreactors are used for large scale production and provide optimum conditions for multiplication of microorganisms.

200. Answer (4)

Bt toxin gene has been expressed in plants in order to provide resistance against lepidopterans, dipterans and coleopterans.

